Watch one program *grow up*

- Cover
- Base
- Handler
- Render
- Updater
- Close

Five steps, one counter. Nothing here replaces anything.



Declare

one value remembers

Grow

lines arrive by morphing

Run

the screen paints the count

The smallest counter that runs

Five lines. Everything after this arrives by morphing, never by replacing the block.

- Cover
- Base
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JAVASCRIPT

step 1 of 4

```
1  const [count, setCount] = useState(0)
2
3  function Counter() {
4    return count
5  }
```

Five lines. The base everything builds on.

Growth pushes the old lines down

The handler lands in the middle. Nothing above it moves; everything below it glides.

- Cover
- Base
- Handler
- Render
- Updater
- Close

JAVASCRIPT

step 2 of 4

```
1  const [count, setCount] = useState(0)
2
3  function inc() {
4    setCount(count + 1)
5  }
6
7  function Counter() {
8    return count
9  }
```

Insert in the middle and the lines below travel.

The render arrives inside the same frame

Two lines fade in, the closing brace glides down, and the frame grows to hold them.

- Cover
- Base
- Handler
- **Render**
- Updater
- Close

JAVASCRIPT

step 3 of 4

```
1  const [count, setCount] = useState(0)
2
3  function inc() {
4    setCount(count + 1)
5  }
6
7  function Counter() {
8    const label = `clicked ${count} times`
9    return <button onClick={inc}>{label}</button>
10 }
```

The frame grows with the lines.

The updater form changes nothing about the shape

Same ten rows. One line fades from stale to updater while nine hold still.

- Cover
- Base
- Handler
- Render
- Updater
- Close

JAVASCRIPT

step 4 of 4

```
1  const [count, setCount] = useState(0)
2
3  function inc() {
4    setCount((c) => c + 1)
5  }
6
7  function Counter() {
8    const label = `clicked ${count} times`
9    return <button onClick={inc}>{label}</button>
10 }
```

reads the queue, never the snapshot

one updater per call, in order

same rows, new reads

Same rows, new reads: the updater form.

Cover
Base
Handler
Render
Updater
Close

Start with the base. Grow it line by line.

5 lines
the base that runs

10 lines
the program grown up

4 steps
each one a morph, none a swap